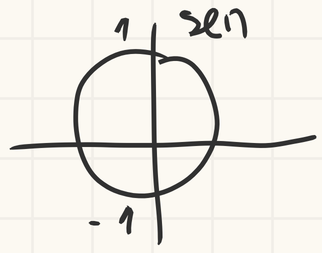
Ficha 1



a)

b) $f(x) = \frac{\pi}{2} - \frac{2\arcsen(1-x)}{3}$

$D_f = \{x \in \mathbb{R} : -1 \leq 1-x \leq 1\}$ C A $1-x \leq 1 \Leftrightarrow -x \leq 1-1 \Leftrightarrow x \geq 0$

$1-x \geq -1 \Leftrightarrow -x \geq -2 \Leftrightarrow x \leq 2$

$D_f =]0, 2] = CD_{f^{-1}}$

$CD_f)$ $-\frac{\pi}{2} \leq \arcsen(x) \leq \frac{\pi}{2} \Leftrightarrow -\frac{\pi}{2} \leq \arcsen(1-x) \leq \frac{\pi}{2} \Leftrightarrow$

$\Leftrightarrow -\frac{\pi}{3} \leq 2\arcsen(1-x) \leq \frac{\pi}{3} \Leftrightarrow \frac{\pi}{2} + \frac{\pi}{3} \geq \frac{\pi}{2} - 2\arcsen \geq \frac{\pi}{2} - \frac{\pi}{3}$

$\Leftrightarrow \frac{3\pi}{6} + \frac{2\pi}{6} \geq \frac{\pi}{2} - 2\arcsen \geq \frac{3\pi}{6} - \frac{2\pi}{6} \Leftrightarrow \frac{5\pi}{6} \geq \frac{\pi}{2} - 2\arcsen \geq \frac{\pi}{6}$

$CD_f = \left[\frac{\pi}{6}, \frac{5\pi}{6}\right] = D_{f^{-1}}$

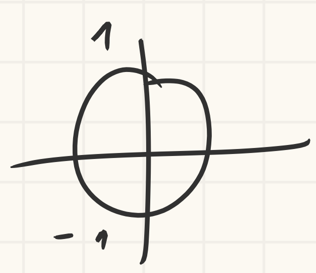
$f^{-1})$ $\frac{\pi}{2} - \frac{2\arcsen(1-x)}{3} = y \Leftrightarrow -\frac{2\arcsen(1-x)}{3} = y - \frac{\pi}{2} \Leftrightarrow$

$\Leftrightarrow \arcsen(1-x) = \left(y - \frac{\pi}{2}\right) \cdot \frac{3}{-2} \Leftrightarrow 1-x = \sen\left(\frac{3y - \frac{3\pi}{2}}{-2}\right)$

$\Leftrightarrow x = -\sen\left(-\frac{3y}{2} + \frac{3\pi}{4}\right) + 1$

$f^{-1} = 1 - \sen\left(\frac{3\pi}{4} - \frac{3y}{2}\right)$

d) $f(x) = e^{\arcsen x}$



$D_f = \{x \in \mathbb{R} : -1 \leq x \leq 1\}$ $D_f = [-1, 1] = CD_{f^{-1}}$

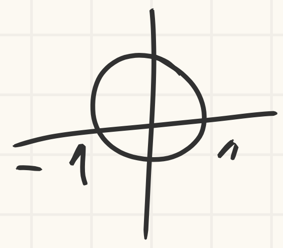
$CD_f)$ $-\frac{\pi}{2} \leq \arcsen x \leq \frac{\pi}{2} \Leftrightarrow e^{-\frac{\pi}{2}} \leq e^{\arcsen x} \leq e^{\frac{\pi}{2}}$

$CD_f = \left[e^{-\frac{\pi}{2}}, e^{\frac{\pi}{2}}\right]$

$f^{-1})\quad e^{\arcsin x} = y \Leftrightarrow \arcsin x = \ln y \Leftrightarrow$

$\Leftrightarrow x = \operatorname{sen}(\ln y)$

$f^{-1} = \operatorname{sen}(\ln(x))$



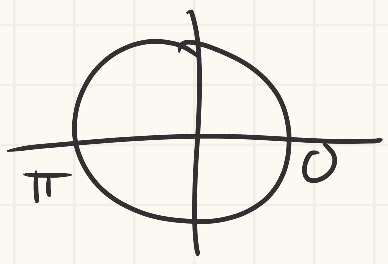
$f)\quad f(x) = 3\arccos(\sqrt{x+4}) - \frac{\pi}{2}$

$D_f = \{ x \in \mathbb{R} : 0 \leq x+4 \wedge -1 \leq \sqrt{x+4} \leq 1 \}$

CA

- $x+4 \geq 0 \Leftrightarrow x \geq -4$
- $\sqrt{x+4} \geq -1$ cod. universal
- $\sqrt{x+4} \leq 1 \Leftrightarrow x+4 \leq 1 \Leftrightarrow x \leq -3$

$D_f = [-4, -3] = CD_{f^{-1}}$



$CD_f)$

$0 \leq \arccos(x) \leq \pi \Leftrightarrow 0 \leq \arccos(\sqrt{x+4}) \leq \frac{\pi}{2} \Leftrightarrow$

$\Leftrightarrow 0 \leq 3\arccos(\sqrt{x+4}) \leq \frac{3\pi}{2} \Leftrightarrow -\frac{\pi}{2} \leq 3\arccos(\sqrt{x+4}) - \frac{\pi}{2} \leq \pi$

$CD_f = [-\frac{\pi}{2}, \pi] = D_{f^{-1}}$

$f^{-1})\quad 3\arccos(\sqrt{x+4}) - \frac{\pi}{2} = y \Leftrightarrow 3\arccos(\sqrt{x+4}) = y + \frac{\pi}{2}$

$\Leftrightarrow \arccos(\sqrt{x+4}) = \frac{y}{3} + \frac{\pi}{6} \Leftrightarrow \sqrt{x+4} = \cos\left(\frac{y}{3} + \frac{\pi}{6}\right)$

$\Leftrightarrow x+4 = \cos^2\left(\frac{y}{3} + \frac{\pi}{6}\right) \Leftrightarrow x = \cos^2\left(\frac{y}{3} + \frac{\pi}{6}\right) - 4$

$g)\quad f(x) = \dfrac{1}{\pi + \arccos(x-2)}$

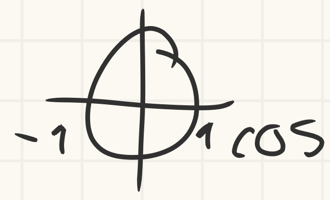
$D_f = \{ x \in \mathbb{R} : -1 \leq x-2 \leq 1 \wedge \pi + \arccos(x-2) \neq 0 \}$

C.A

- $x-2 \geq -1 \iff x \geq 1$
- $x-2 \leq 1 \iff x \leq 3$
- $\pi + \arccos(x-2) \neq 0 \iff \arccos(x-2) \neq -\pi \iff x-2 \neq \cos(-\pi)$

$\iff x \neq \cos(-\pi) - 2 \iff x \neq -3$

$$D_f = [1, 3] = CD_{f^{-1}}$$



$CD_f)$ $0 \leq \arccos(x-2) \leq \pi \iff \pi \leq \pi + \arccos(x-2) \leq 2\pi$

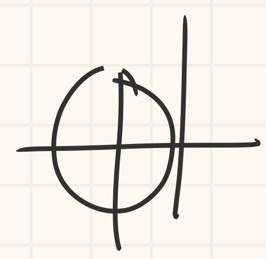
$\iff \dfrac{1}{\pi} \geq \dfrac{1}{\pi + \arccos(x-2)} \geq \dfrac{1}{2\pi}$

$$CD_f = \left[\frac{1}{2\pi}, \frac{1}{\pi}\right]$$

$f^{-1})$ $y = \dfrac{1}{\pi + \arccos(x-2)} \iff \dfrac{1}{y} = \pi + \arccos(x-2) \iff$

$\iff \dfrac{1}{y} - \pi = \arccos(x-2) \iff \cos\left(\dfrac{1}{y} - \pi\right) = x-2$

$\iff \cos\left(\dfrac{1}{y} - \pi\right) + 2 = x$ $\qquad f^{-1} = \cos\left(\dfrac{1}{x} - \pi\right) + 2$



h) $f(x) = \pi - 3\operatorname{arctg}\left(\dfrac{x-1}{2}\right)$

$D_f = \{x \in \mathbb{R}\} = CD_{f^{-1}}$

$CD_f)$ $-\dfrac{\pi}{2} \leq \operatorname{arctg}\left(\dfrac{x-1}{2}\right) \leq \dfrac{\pi}{2} \iff -\dfrac{3\pi}{2} \leq 3\operatorname{arctg}\left(\dfrac{x-1}{2}\right) \leq \dfrac{3\pi}{2}$

$\iff \pi + \dfrac{3\pi}{2} \geq \pi - 3\operatorname{arctg}\left(\dfrac{x-1}{2}\right) \geq \pi - \dfrac{3\pi}{2} \iff$

$\iff \dfrac{5\pi}{2} \geq \pi - 3\operatorname{arctg}\left(\dfrac{x-1}{2}\right) \geq -\dfrac{\pi}{2}$

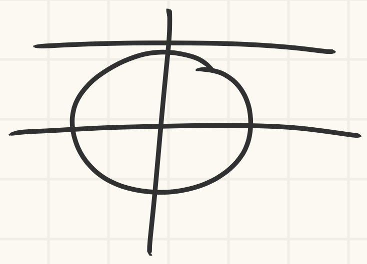
$$CD_f = \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right] = D_{f^{-1}}$$

$f^{-1})$ $\pi - 3\operatorname{arctg}\left(\frac{x-1}{2}\right) = y \Leftrightarrow -3\operatorname{arctg}\left(\frac{x-1}{2}\right) = y - \pi$

$\Leftrightarrow \operatorname{arctg}\left(\frac{x-1}{2}\right) = \frac{y-\pi}{-3} \Leftrightarrow \frac{x-1}{2} = \operatorname{tg}\left(\frac{y-\pi}{-3}\right)$

$\Leftrightarrow x - 1 = 2\operatorname{tg}\left(\frac{-y+\pi}{3}\right) \Leftrightarrow x = 2\operatorname{tg}\left(\frac{-y+\pi}{3}\right) + 1$

$$f^{-1} = 2\operatorname{tg}\left(\frac{-x+\pi}{3}\right) + 1$$



i) $f(x) = \operatorname{arccotg}(\ln(x+1))$

$D_f = \{ x \in \mathbb{R} : x + 1 > 0 \}$ $\quad D_f =]-1, +\infty[\; = CD_{f^{-1}}$

$CD_f)$

$0 < \operatorname{arccotg}(x) < \pi \Leftrightarrow 0 < \operatorname{arccotg}(\ln(x)) < \pi$

$CD_f =]0, \pi[\; = D_{f^{-1}}$

$f^{-1})$ $\operatorname{arccotg}((\ln(x+1)) = y \Leftrightarrow \ln(x+1) = \operatorname{cotg}(y) \Leftrightarrow x+1 = e^{\operatorname{cotg}(y)}$

$\Leftrightarrow x = e^{\operatorname{cotg}(y)} - 1$ $\quad f^{-1}(x) = e^{\operatorname{cotg}(x)} - 1$

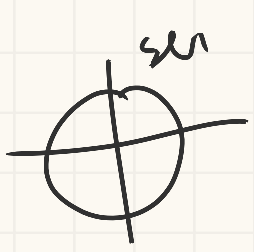
7) $f(x) = \operatorname{arcsen}(x^2 - 1)$

$D_f = \{ x \in \mathbb{R} : -1 \leq x^2 - 1 \leq 1 \}$

C.A

- $x^2 - 1 \geq -1 \Leftrightarrow x^2 \geq 0$. cond univ.
- $x^2 - 1 \leq 1 \Leftrightarrow x^2 \leq 2 \Leftrightarrow x \leq \pm\sqrt{2}$

$\Leftrightarrow x \geq -\sqrt{2} \wedge x \leq \sqrt{2}$

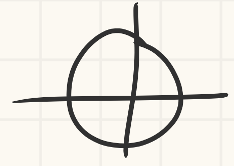


$D_f = [-\sqrt{2}, \sqrt{2}]$

$CD_f)$ $\quad -\frac{\pi}{2} \leq \operatorname{arcsen}(x) \leq \frac{\pi}{2} \Leftrightarrow 0 \leq \operatorname{arcsen}(x^2) \leq \frac{\pi}{2}$

$\Leftrightarrow -\frac{\pi}{2} \leq \operatorname{arcsen}(x^2-1) \leq \frac{\pi}{2}$ $\quad CD_f = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Interseção de f com os eixos?



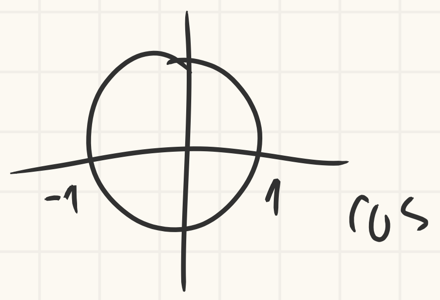
$f(0) = \arcsin(0^2 - 1) = \arcsin(-1) = -\frac{\pi}{2}$ intersecta em $y = -\frac{\pi}{2}$

$f(x) = 0 \Leftrightarrow \arcsin(x^2 - 1) = 0 \Leftrightarrow x^2 - 1 = \operatorname{sen}(0) \Leftrightarrow x^2 = 0 + 1 \Leftrightarrow x^2 = 1$

intersecta em $f(1)$ e $f(-1)$

em $(-1, 0)$, $(1, 0)$ e $(0, -\frac{\pi}{2})$

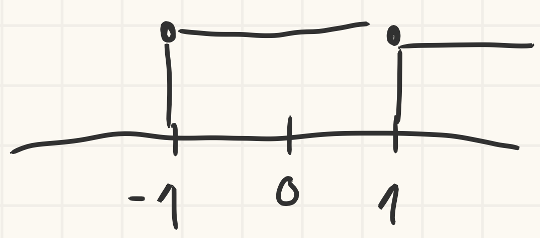
8) $g(x) = \arccos\left(\frac{1}{x}\right)$, D_g, CD_g e zeros?



$D_g = \{x \in \mathbb{R} : -1 \leq \frac{1}{x} \leq 1 \wedge x \neq 0\}$

C.A

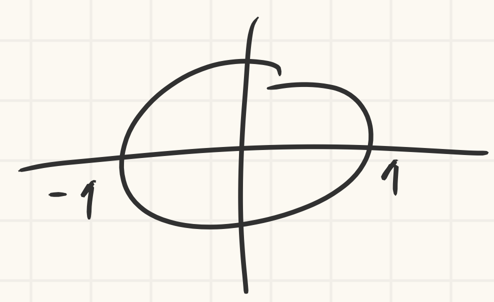
- $\frac{1}{x} \geq -1 \Leftrightarrow x \leq -1 \wedge x > 0$ $\quad]-\infty, -1] \cup [1, +\infty[$
- $\frac{1}{x} \leq 1 \Leftrightarrow x \geq 1 \wedge x < 0$



como $D_f =]-\infty, -1] \cup [1, +\infty[$ nunca vai ser 0

$CD_g)$ $\quad 0 \leq \arccos \leq \pi \Leftrightarrow$ $\arccos\left(\frac{1}{x}\right)$

$CD_g = [0, \pi] / \{\frac{\pi}{2}\}$



zeros) $\quad g(x) = 0 \Leftrightarrow \arccos\left(\frac{1}{x}\right) = 0 \Leftrightarrow \frac{1}{x} = \cos(0) \Leftrightarrow \frac{1}{x} = 1 \Leftrightarrow x = 1$

10) $g(x) = \begin{cases} \sqrt{-x} \cdot \operatorname{sen}\left(\frac{1}{x^3}\right), & x < 0 \\ \operatorname{arctg}(x^2), & x \geq 0 \end{cases}$

$\lim_{x \to 0} g(x) = ?$

$$\lim_{x\to 0^+} \operatorname{arctg}(0^?) = 0$$

$$\lim_{x\to 0^-} g(x) = \lim_{x\to 0^-} \sqrt{-x} \cdot \operatorname{sen}\left(\frac{1}{x^3}\right) = 0 \cdot \lim_{x\to 0^-} \operatorname{sen}\left(\frac{1}{x^3}\right) = 0$$ ✓

13- $x^5 - 3x = 1$ admite pelo menos uma solução em $]1,2[$?

$f(x)\ x^5 - 3x - 1 = 0$

1) $1 - 3 - 1 = -3$

2) $32 - 6 - 1 = 32 - 7$

$f(1) \cdot f(2) < 0$ então admite um zero em $]1, 2[$

17) $f(x) = \begin{cases} 1 - x & \text{se} \quad x < 0 \\ x^2 + 3 & \text{se} \quad x \geq 0 \end{cases}$ tem mínimo global em $[-1, 1]$?

$f(-1) = 1 - (-1) = 2$

$\lim_{x\to -1} f(x) = 2$

$f(1) = 1 + 3 = 4$

$\lim f(1) = 4$

$f(0) = 3$

$\lim_{x\to 0^+} = 3$

$\lim_{x\to 0^-} = 1$

22) $f(x) = (x - 1)(x^2 + 3x) \qquad f'(x) = ?$

$$f'(x) = (x-1)' \cdot (x^2 + 3x) + (x^2 + 3x)'(x - 1) =$$
$$= 1 \cdot (x^2 + 3x) + (2x + 3)(x - 1) = x^2 + 3x + 2x^2 - 2x + 3x - 3$$
$$= 2x^2 + x^2 + 3x - 2x + 3x - 3 = 3x^2 + 4x - 3$$

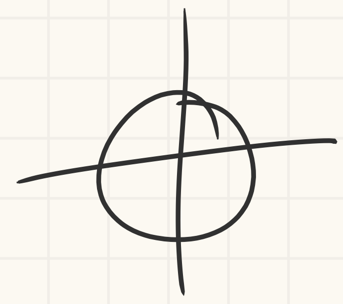
c) $f(x) = \dfrac{\cos x}{1 - \operatorname{sen} x} \qquad f'(x) = ?$

$$f'(x) = \frac{\cos' x\,(1 - \operatorname{sen} x) - \cos x\,(1 - \operatorname{sen} x)'}{(1 - \operatorname{sen} x)^2} =$$

$$= \frac{-\operatorname{sen} x\,(1 - \operatorname{sen} x) - \cos x\,(-\cos x)}{(1 - \operatorname{sen} x)^2} =$$

$$= \frac{-\operatorname{sen} x + \operatorname{sen}^2 x + \cos^2 x}{1 - 2\operatorname{sen} x + \operatorname{sen}^2 x} = \frac{-\operatorname{sen} x + 1}{(1 - \operatorname{sen} x)^2} =$$

$$= \frac{1}{1 - \operatorname{sen} x}$$



33) $f(x) = \operatorname{arcsen}(1-x) + \sqrt{2x - x^2}$

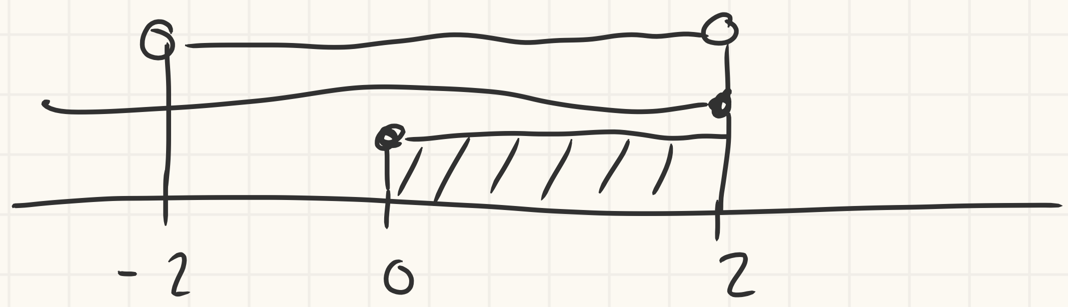
a) $D_f = ?$

$D_f = \{ x \in \mathbb{R} : 2x - x^2 > 0 \ \wedge \ -1 \leq 1 - x \leq 1$

CA

- $2x - x^2 > 0 \Leftrightarrow 2x > x^2 \Leftrightarrow x \in]-2, 2[$
- $1 - x \geq -1 \Leftrightarrow -x \geq -2 \Leftrightarrow x \leq 2$
- $1 - x \leq 1 \Leftrightarrow -x \leq 0 \Leftrightarrow x \geq 0$

$D_f = [0, 2]$



b) $f'(x) = -\dfrac{x}{\sqrt{2x - x^2}}$?

$f'(x) = \operatorname{arcsen}'(1-x) + \left(\sqrt{2x - x^2}\right)' =$

$$= \frac{-1}{\sqrt{1-(1-x)^2}} + \frac{1}{2}(2x - x^2)^{-\frac{1}{2}} \cdot (2 - 2x)$$

$$= -\frac{1}{\sqrt{2x - x^2}} + \frac{1}{2\sqrt{2x - x^2}}(2 - 2x) = \frac{-2}{2\sqrt{2x-x^2}} + \frac{2 - 2x}{2\sqrt{2x - x^2}}$$

$$= \frac{-2 + 2 - 2x}{2\sqrt{2x - x^2}} = \frac{-2x}{2\sqrt{2x - x^2}} =$$

$$= -\frac{x}{\sqrt{2x - x^2}} \checkmark$$

CA $1 - (1-x)^2 = 1 - (1^2 - 2x + x^2)$

$= +2x - x^2$

c) Max e min globais ?

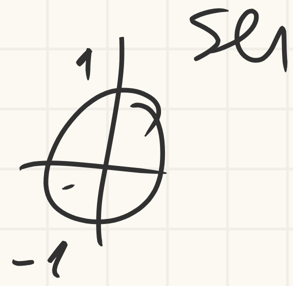
Como f é contínua em $[0, 2]$, ela atinge max e mínimo globais nesse intervalo (T. Weierstrass)

Como $f'(x) < 0$ em $]0,2[\rightarrow f$ é estrita/ decrescente

então maximizante $= f(0)$ e minimizante $= f(2)$

máximo) $f(0) = \arcsin(1-0) + \sqrt{2.0 - 0^2} = \arcsin(1) + 0$

$f(0) = \frac{\pi}{2}$



mínimo) $f(2) = \arcsin(1-2) + \sqrt{2.2 - 2^2} = \arcsin(-1) + \sqrt{0}$

$= -\frac{\pi}{2} + 0$

- mínimo global $-\frac{\pi}{2}$ em $f(2)$
- máximo global $\frac{\pi}{2}$ em $f(0)$

d) CD_f ? $CD_f = [-\frac{\pi}{2}, \frac{\pi}{2}]$

34) $h(x) = \operatorname{arctg}(x^2 - 4x)$ Extremos e monotonia?

$Dh = \{x \in \mathbb{R}\}$

$h'(x) = \dfrac{2x-4}{1+(x^2-4x)^2}$

$h'(x) = 0 \Leftrightarrow \dfrac{2x-4}{1+(x^2-4x)^2} = 0 \Leftrightarrow 2x-4 = 0 \land 1+(x^2-4x)^2 \neq 0$

cond. universal

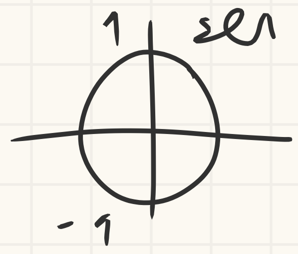
$2x - 4 = 0 \Leftrightarrow 2x = 4 \Leftrightarrow x = 2$

	$-\infty$	2	$+\infty$
$h'(x)$	$-$	0	$+$
$h(x)$	↘	min	↗

h é estritamente decrescente em $]-\infty, 2[$ e est. cresc. em $]2, +\infty[$

2 é minimizante global de h

$h(2) = \text{arctg}(2^2 - 4 \cdot 2) = \text{arctg}(-4)$ é o mínimo global de h



35) $g(x) = \text{arcsen}((x-1)^2)$

a) $D_g = ?$ $\quad D_g = \{x \in \mathbb{R} : -1 \leq (x-1)^2 \leq 1\}$

C.A

- $(x-1)^2 \geq -1$ cond. univ $\qquad D_g = [0, 2]$
- $(x-1)^2 \leq 1 \Leftrightarrow x - 1 \leq 1^2 \rightarrow x \leq +1+1 \Leftrightarrow x \leq 2$

  $x \leq -1+1 \Leftrightarrow x \geq 0$

c) Extremos e monotonia?

$$g'(x) = \frac{(2x-2)\cdot 1}{\sqrt{1-(x-1)^4}} = \frac{2x-2}{\sqrt{1-(x-1)^4}}$$

$$g'(x) = 0 \Leftrightarrow \frac{2x-2}{\sqrt{1-(x-1)^4}} = 0 \Leftrightarrow 2x-2=0 \wedge 1-(x-1)^4 > 0$$

C.A

- $2x - 2 = 0 \Leftrightarrow 2x = 2 \Leftrightarrow x = 1$
- $1-(x-1)^4 > 0 \Leftrightarrow 1 > (x-1)^4$

  $\Leftrightarrow |x-1| < 1 \Leftrightarrow x - 1 < \pm 1$

  $\Leftrightarrow x - 1 < 1 \wedge x - 1 > -1$

  $\Leftrightarrow x < 2 \wedge x > 0 \rightarrow x \in]0, 2[$

	0		1		2
g'		$-$	0	$+$	
g	$\frac{\pi}{2}$	↘	0	↗	$\frac{\pi}{2}$

$g(0) = \text{arcsen}((0-1)^2) = \text{arcsen}(1) = \frac{\pi}{2}$

$g(1) = \text{arcsen}(0) = 0$

$g(2) = \text{arcsen}((2-1)^2) = \text{arcsen}(1) = \frac{\pi}{2}$

g é: estritamente decrescente em $[0, 1]$

" crescente em $[1,2]$

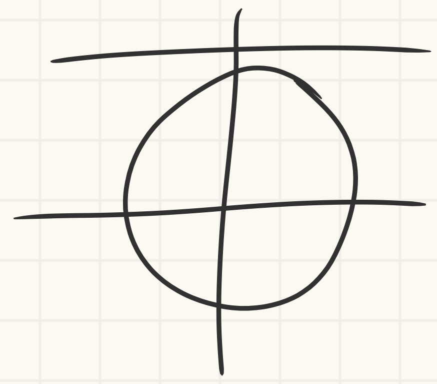
máximo global $= \frac{\pi}{2}$, maximizantes $= 0$ e 2

mínimo global $= 0$, minimizante $= 1$

d) Invertível?

Como $g(0) = g(2) \Rightarrow g$ n é injetiva, logo tb n é invertível

34) $h(x) = \operatorname{arctg}(x^2 - 4x)$ Extremos? Monotonia?



$D_h = \mathbb{R}$

$h'(x) = \operatorname{arctg}(x^2-4x)' = \dfrac{2x-4}{1+(x^2-4x)^2}$

$h'(x) = 0 \Leftrightarrow \dfrac{2x-4}{1+(x^2-4x)^2} = 0 \Leftrightarrow 2x-4 = 0 \wedge \underbrace{1+(x^2-4x)^2 \neq 0}_{\text{cond. univ.}}$

$\Leftrightarrow x = 2$

	$-\infty$	2	$+\infty$
h'	$-$	0	$+$
h	↘		↗

$h(2) = \operatorname{arctg}(2^2 - 4.2) = \operatorname{arctg}(4-8)$
$= \operatorname{arctg}(-4)$

35) $g(x) = \operatorname{arcsen}((x-1)^2)$

a) D_g?

$D_g = \{x \in \mathbb{R} : -1 \leq (x-1)^2 \leq 1\}$

C.A

- $(x-1)^2 \geq -1$ cond. universal
- $(x-1)^2 \leq 1 \Leftrightarrow |x-1| \leq 1 \Leftrightarrow$ $x-1 \leq 1 \Leftrightarrow x \leq 2$; $x-1 \geq -1 \Leftrightarrow x \geq 0$

$D_g = [0, 2]$

b) $g(x) = \frac{\pi}{6}$ tem pelo menos uma solução em $[1,2]$?

$$\arcsin((x-1)^2) = \frac{\pi}{6} \Leftrightarrow \arcsin((x-1)^2) - \frac{\pi}{6} = 0$$

em 1)
$$\arcsin((1-1)^2) - \frac{\pi}{6} \Leftrightarrow \arcsin(0) - \frac{\pi}{6} = -\frac{\pi}{6}$$

em 2)
$$\arcsin((2-1)^2) - \frac{\pi}{6} \Leftrightarrow \arcsin(1) - \frac{\pi}{6} = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi}{6} - \frac{\pi}{6}$$
$$= \frac{2\pi}{6} = \frac{\pi}{3}$$

Pelo T. Bolzano como $g(1) - \frac{\pi}{6} \cdot g(2) - \frac{\pi}{6} < 0$
então $g(x) = \frac{\pi}{6}$ tem pelo menos uma solução em $[1,2]$

c) Extremos e monotonia?

$$g'(x) = \arcsin((x-1)^2)' = \frac{((x-1)^2)'}{\sqrt{1-(x-1)^4}} = \frac{2(x-1).(x-1)'}{\sqrt{1-(x-1)^4}}$$

$$= \frac{2(x-1)}{\sqrt{1-(x-1)^4}}$$

$$g'(x) = 0 \Leftrightarrow \frac{2(x-1)}{\sqrt{1-(x-1)^4}} = 0 \Leftrightarrow x-1 = 0 \wedge 1-(x-1)^4 > 0$$

$$\Leftrightarrow x = 1 \wedge -(x-1)^4 > -1 \Leftrightarrow x = 1 \wedge (x-1)^4 < 1$$

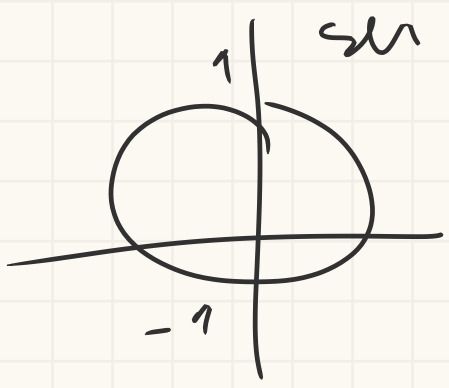
$$|x-1| < 1$$

$x - 1 < 1 \Leftrightarrow x < 2$

$x - 1 > -1 \Leftrightarrow x > 0$

$x \in]0,2[$

	0		1		2
g'		$-$	0	$+$	
g	$\frac{\pi}{2}$	↘	0	↗	$\frac{\pi}{2}$



$$g(0) = \arcsin((0-1)^2) = \frac{\pi}{2}$$

$$g(1) = \arcsin(0) = 0$$

$g(2) = \arcsin(1) = \frac{\pi}{2}$

d) Inventível? Não pq $g(0) = g(2) \to g$ n é injetiva $\to g$ n é inventível

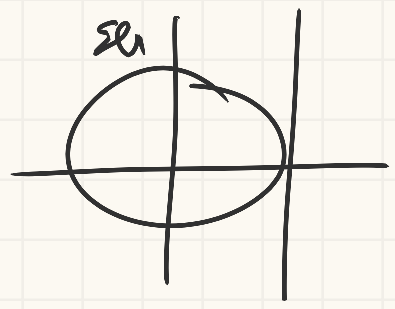
36

a) $\lim_{x \to +\infty} (\ln(3x^2+2) - \ln(x^2)) \overset{\infty - \infty}{=} \lim \left(\ln\left(\frac{3x^2+2}{x^2}\right)\right) = \lim\left(\ln\left(\frac{3x^2}{x^2} + \frac{2}{x^2}\right)\right)$

$= \lim\left(\ln\left(3 + \frac{2}{x^2}\right)\right) = \ln(3+0) = \ln(3)$

38) $g(x) = \begin{cases} \text{arctg}(x^2) & , \ x \leq 0 \\ x^2 \operatorname{sen}\left(\frac{1}{x}\right) & , \ x > 0 \end{cases}$

a) diferenciavel em $x = 0$? $g'(0) = ?$



$g(0) = \text{arctg}(0) = 0$

$\lim_{x \to 0^-} g(x) = \text{arctg}(0) = 0$

$\lim_{x \to 0^+} g(x) = \lim_{x \to 0^+} x^2 \operatorname{sen}\left(\frac{1}{x}\right) \overset{0.0}{=} 0$

42)

a) $T_0^3 (x^3 + 2x + 1) = ? \ = \sum_{k=0}^{3} \frac{f^{(k)}(0)}{k!} (x-0)^k$

$f(x) = x^3 + 2x + 1$ $\qquad f(0) = 1$

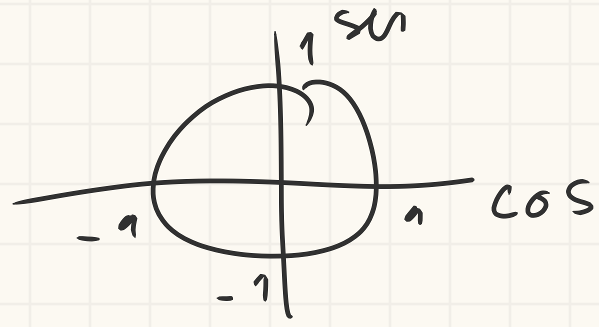
$f'(x) = 3x^2 + 2$ $\qquad f'(0) = 2$

$f''(x) = 6x$ $\qquad f''(0) = 0$

$f'''(x) = 6$ $\qquad f'''(0) = 6$

$= 1 + 2(x-0) + 0 + \frac{6}{3!}(x-0)^3 = 1 + 2x + x^3$

b) $T_{\pi}^{3}(\cos x) = ?$



$f(x) = \cos x \qquad f(\pi) = \cos(\pi) = -1$

$f'(x) = -\operatorname{sen} x \qquad f'(\pi) = -\operatorname{sen}(\pi) = 0$

$f''(x) = -\cos x \qquad f''(\pi) = -\cos(\pi) = 1$

$f'''(x) = \operatorname{sen} x \qquad f'''(\pi) = \operatorname{sen}(\pi) = 0$

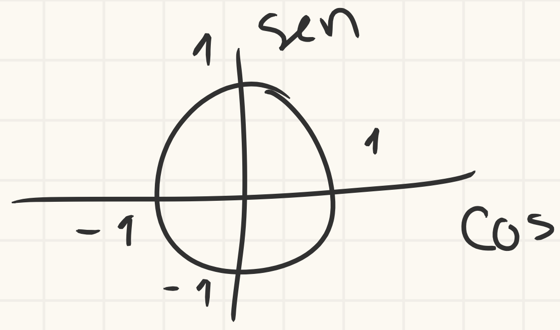
$$T_{\pi}^{3} = -1 + \frac{0}{1}(x-\pi)^1 + \frac{1}{2!}(x-\pi)^2 + \frac{0}{3!}(x-\pi)^3$$

$$= -1 + 0 + \frac{1}{2}(x-\pi)^2 + 0 = \frac{(x-\pi)^2}{2} - 1$$

Ex miniteste tipo:

5. Usando o polinómio de MacLaurin de ordem 2 de $f(x) = \cos x$, $T_0^2(f(x))$, podemos concluir que um valor aproximado de $\cos(\frac{1}{2})$ é igual a:

$T_0^2(\cos x) = ?$



$f(x) = \cos x \qquad f(0) = \cos 0 = 1$

$f'(x) = -\operatorname{sen} x \qquad f'(0) = -\operatorname{sen} 0 = 0$

$f''(x) = -\cos x \qquad f''(0) = -\cos 0 = -1$

$$T_0^2(\cos x) = 1 + 0 - \frac{1}{2!}(x-0)^2 = 1 - \frac{x^2}{2}$$

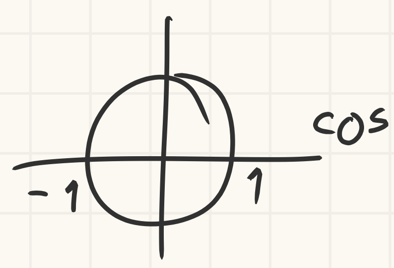
$$x = \tfrac{1}{2} \to 1 - \frac{(1/2)^2}{2} = 1 - \frac{1/4}{2} = 1 - \frac{1}{4} \cdot \frac{1}{2} = 1 - \frac{1}{8} = \frac{7}{8}$$

47) zeros de $g(x) = \begin{cases} \arccos(x^2) & , \quad -1 \leq x < 0 \\ e^{-x+1} & , \quad x \geq 0 \end{cases}$

$D_g = \{ x \in \mathbb{R} : x^2 \leq 1 \} \to [-1, 1]$

C.A $\quad x^2 \leq 1 \Leftrightarrow x \leq 1 \wedge x \geq -1$

$\arccos(x^2) = 0 \Leftrightarrow x^2 = \cos 0 \Leftrightarrow x^2 = 1 \Leftrightarrow \cancel{x = 1} \wedge x = -1$



$e^{-x+1} = 0 \quad \Leftrightarrow$
imp.

44) $f(x) = e^x$

a) $T_0^n = ?$ $\quad T_0^n = 1 + 1.(x-0)^1 + \frac{1}{2!}(x-0)^2 + \dots + \frac{x^n}{n!} + \frac{e^\theta}{(n+1)!} x^{n+1}, \ \theta \in [0, x]$

$f(x) = e^x \quad f(0) = 1$

$f'(x) = e^x \quad f'(0) =$

48) $h(x) = \arccos(e^{x-1}) + \pi$, invertível? $h^{-1} = ?$ $Dh^{-1} = ?$ $CDh^{-1} = ?$

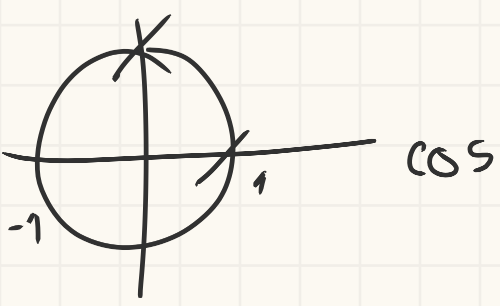
$Dh = \{ x \in \mathbb{R} : -1 \leq e^{x-1} \leq 1 \}$ $\qquad Dh =]-\infty, 1] = CDh^{-1}$

CA

- $e^{x-1} \geq -1$ cond. univ
- $e^{x-1} \leq 1 \Leftrightarrow x - 1 \leq \ln(1) \Leftrightarrow x - 1 \leq 0 \Leftrightarrow x \leq 1$

$e^{-\infty}$ a e^0 = 0 a 1

CDh) $\quad 0 \leq \arccos(x) \leq \pi \Leftrightarrow 0 \leq \arccos(e^{x-1}) \leq \frac{\pi}{2} \Leftrightarrow$



$\Leftrightarrow \pi \leq \arccos(e^{x-1}) + \pi \leq \frac{3\pi}{2}$

$CDh = [\pi, \frac{3\pi}{2}] = Dh^{-1}$

h^{-1}) $\arccos(e^{x-1}) + \pi = y \Leftrightarrow e^{x-1} = \cos(y - \pi) \Leftrightarrow x - 1 = \ln(\cos(y - \pi))$

$\Leftrightarrow x = \ln(\cos(y - \pi)) + 1$

$h^{-1}(x) = \ln(\cos(x - \pi)) + 1$

50) $4x^3 - 6x^2 + 1 = 0$ tem 3 raizes em que intervalos?

$(4x^3 - 6x^2 + 1)' = 12x^2 - 12x$

entre $]0,1[$ tem no max um zero

$12x^2 - 12x = 0 \Leftrightarrow 12x^2 = 12x \Leftrightarrow x^2 = x \Leftrightarrow x = 0 \wedge x = 1$

em $x = 0$) $\quad 4.0 - 6.0 + 1 = 1$

em $x = 1$) $\quad 4 - 6 + 1 = -1$

em $x = -1$) $\quad 4.(-1) - 6 + 1 = -4 - 6 + 1 = 9$

	$-\infty$	-1		0		1	$+\infty$
g'	$-$	0	$-$	0	$-$	0	$+$
g	↘	9	↘	1	↘	-1	↗

$12x^2 = 12x \Leftrightarrow 12x = 12 \Leftrightarrow$